

mTLS Error-to-Fix Mapping

This document maps common mTLS setup and deployment errors to their root causes and remediation steps, based on real failures observed during user studies.

Client-Side Setup Errors

Error 1: “UNABLE_TO_LOAD_CONFIG”

When it occurs:

```
openssl req -new -key client.key -out client.csr -config csr.conf
```

Root cause: CSR configuration file not found or in wrong location

Remediation: 1. Verify `csr.conf` exists in current directory: `ls -la csr.conf` 2. If in different directory, use full path: `openssl req ... -config /path/to/csr.conf` 3. Check file permissions: `chmod 644 csr.conf`

Error 2: “req_unable_to_create_request”

When it occurs: Configuration file exists but has syntax errors

Root cause: Malformed CSR config (missing sections, bad syntax, etc.)

Remediation: 1. Validate syntax: `openssl req -in csr.conf -noout -text` 2. Check for required sections: `[req]`, `[req_distinguished_name]`, `[v3_req]` 3. Ensure no trailing spaces or tabs 4. Compare against sample-`csr.conf`

Error 3: “CSR rejected: Missing Subject Alternative Name”

When it occurs: CA rejects CSR during submission

Root cause: CSR missing SAN (Subject Alternative Name) extension

Remediation: 1. Verify CSR has SAN: `openssl req -in client.csr -noout -text | grep "Subject Alternative Name"` 2. If missing, regenerate CSR with config that includes SAN section:

```
[v3_req]
subjectAltName = DNS:localhost, DNS:example.com
```

3. Recreate CSR: `openssl req -new -key client.key -out client.csr -config csr.conf -extensions v3_req`

Error 4: “MAC verification failed”

When it occurs: Importing PKCS#12 file into browser/OS

Root cause: Wrong password entered during import

Remediation: 1. Verify password used during export: `openssl pkcs12 -export -out client.p12` 2. Re-export with known password:

```
openssl pkcs12 -export -out client.p12 \  
-inkey client.key -in client.crt \  
-passout pass:MyPassword123
```

3. Try importing again with correct password

Error 5: “Certificate not found in browser”

When it occurs: Browser doesn’t prompt for certificate; connects without mTLS

Root cause: Certificate not imported to correct OS/browser store

Remediation: 1. **macOS:**

```
security find-identity -v -p sslclient
```

Should list your cert. If not, re-import via Keychain.

2. **Windows:**

```
Get-ChildItem -Path Cert:\CurrentUser\My
```

Should show cert. If not, re-import via certmgr.msc

3. **Linux (Firefox):**

```
certutil -L -d ~/.mozilla/firefox/YOUR_PROFILE
```

Should list cert. If not, re-import via Firefox Preferences

4. **Restart browser completely** — critical step often missed

Error 6: “Bad file descriptor” or “No such file or directory”

When it occurs: Running OpenSSL commands

Root cause: File paths incorrect or files don’t exist

Remediation: 1. Verify all files exist:

```
ls -la client.key client.csr ca.crt
```

2. Use absolute paths if relative paths fail:

```
openssl pkcs12 -export -out /full/path/client.p12 \  
-inkey /full/path/client.key -in /full/path/client.crt
```

3. Check file permissions: `chmod 600 client.key`

Certificate Authority Errors

Error 7: “CA cert not recognized”

When it occurs: During CSR submission to CA portal

Root cause: CSR signature invalid or CA configuration incompatible

Remediation: 1. Verify CSR signature: `openssl req -in client.csr -noout -verify` 2. Check CSR matches requirements: `openssl req -in client.csr -noout -text` 3. Regenerate CSR with correct config 4. Contact CA with CSR for validation

Error 8: “Certificate chain incomplete”

When it occurs: Server certificate installation fails

Root cause: Server cert issued by intermediate CA; chain not provided

Remediation: 1. Request full certificate chain from CA: - Root CA cert - Intermediate CA cert (if applicable) - Server cert - CA bundle file

2. Install bundle in correct order:

```
cat server.crt intermediate.crt root.crt > chain.crt
```

3. Configure nginx:

```
ssl_certificate chain.crt;  
ssl_certificate_key server.key;
```

Server-Side Deployment Errors

Error 9: “SSL: CERTIFICATE_VERIFY_FAILED”

When it occurs: Client connects with curl or browser

Root cause: Client certificate not signed by trusted CA, or CA cert not configured on server

Remediation: 1. Verify CA cert is correct:

```
openssl x509 -in ca.crt -noout -text
```

2. Ensure nginx has correct CA:

```
ssl_client_certificate /etc/nginx/certs/ca.crt;
```

3. Verify client cert was signed by CA:

```
openssl verify -CAfile ca.crt client.crt
```

4. Check certificate chain:

```
openssl s_client -connect example.test:443 -showcerts
```

Error 10: “ssl_client_verify=FAILED” in logs

When it occurs: nginx logs show failed verification

Root cause: Multiple possible causes (invalid cert, expired cert, signature mismatch, etc.)

Remediation: 1. Check certificate validity:

```
openssl x509 -in client.crt -noout -dates
```

2. Check certificate is signed by CA:

```
openssl verify -CAfile ca.crt client.crt
```

3. Verify nginx config:

```
ssl_verify_client on;  
ssl_verify_depth 2;  
ssl_client_certificate /etc/nginx/certs/ca.crt;
```

4. Check nginx error log: `tail -f /var/log/nginx/error.log`

Error 11: “Client certificate required” (HTTP 403)

When it occurs: Browser or curl receives 403 Forbidden

Root cause: Server requires client cert, but client didn’t send one

Remediation: 1. Verify nginx requires mTLS:

```
ssl_verify_client on;
```

2. Client must provide certificate:

```
curl --cert client.p12:password https://example.test/
```

3. In browser: Certificate must be imported to correct store

4. Verify cert is in correct store:

- macOS: `security find-identity`
- Windows: `certmgr.msc`
- Linux: `certutil -L`

Multi-Device Errors

Error 12: “Cannot transfer certificate to new device”

When it occurs: Trying to export/import PKCS#12 to different machine

Root cause: Private key encrypted; password forgotten or incorrect

Remediation: 1. Verify PKCS#12 integrity:

```
openssl pkcs12 -in client.p12 -noout -info -passin pass:password
```

2. If password forgotten, need to regenerate:

- Request certificate revocation
- Generate new CSR and certificate
- Create new PKCS#12

3. Export with strong, memorable password:

```
openssl pkcs12 -export -out client.p12 \  
-inkey client.key -in client.crt \  
-passout pass:Tr0pic@lThund3r$unset
```

4. Store password securely (password manager recommended)

Error 13: “Certificate not recognized on second device”

When it occurs: Same PKCS#12 file, different device, browser won't use it

Root cause: Certificate not imported to second device's OS/browser store

Remediation: 1. Re-import PKCS#12 on new device following device-specific instructions 2. Verify import: See Error 5 remediation for verification steps 3. Restart browser completely 4. Test: `curl --cert client.p12:password https://example.test/`

Browser Integration Errors

Error 14: “Certificate selection dialog not appearing”

When it occurs: Browser doesn't prompt to select certificate

Root cause: - Certificate not imported - Browser doesn't see certificate in store
- Browser cache issue

Remediation: 1. Verify cert in store (see Error 5) 2. Clear browser cache: Ctrl+Shift+Delete (or Cmd+Shift+Delete on macOS) 3. Restart browser completely 4. Hard refresh: Ctrl+F5 (or Cmd+Shift+R on macOS) 5. Try different browser to isolate the issue

Error 15: “Certificate dialog appears but cert is grayed out/disabled”

When it occurs: Certificate listed but can't be selected

Root cause: Certificate missing Extended Key Usage: Client Authentication

Remediation: 1. Check certificate extensions:

```
openssl x509 -in client.crt -noout -text | grep -A 5 "Extended Key Usage"
```

2. Should show:

```
Extended Key Usage:  
    TLS Web Client Authentication
```

3. If missing, regenerate with correct extensions:

```
openssl req -new -key client.key -out client.csr \
  -addext "extendedKeyUsage = clientAuth"
```

Logging and Diagnostic Errors

Error 16: “Client cert info not appearing in nginx logs”

When it occurs: nginx logs don’t show `ssl_client_s_dn` or `ssl_client_serial`

Root cause: Log format not configured

Remediation: 1. Verify log format includes cert variables:

```
log_format mTLS_log '$ssl_client_s_dn $ssl_client_serial';
```

2. Apply format to access log:

```
access_log /var/log/nginx/access.log mTLS_log;
```

3. Reload nginx: `nginx -s reload`

4. Make request and check log: `tail -f /var/log/nginx/access.log`

Error 17: “Diagnostic endpoint returns ‘No certificate provided’”

When it occurs: Accessing `/tls-info` without client cert

Root cause: Expected behavior — no cert sent with request

Remediation: 1. Send request WITH client certificate:

```
curl --cert client.p12:password https://example.test/tls-info
```

2. If still shows “No certificate”, check:

```
location /tls-info {
    proxy_set_header X-SSL-Client-Cert-DN $ssl_client_s_dn;
}
```

Testing Checklist

Before deploying to production, verify:

- ☐ Client certificate installation works on all target devices
- ☐ Server accepts valid certificates
- ☐ Server rejects invalid certificates
- ☐ Server rejects requests without certificates
- ☐ nginx logs capture certificate information
- ☐ Diagnostic endpoints work correctly
- ☐ Certificate renewal process documented
- ☐ Error responses are user-friendly
- ☐ Monitoring/alerts configured for failed verifications

Contact and Support

If an error isn't listed here:

1. Check complete error output: `curl -v` or browser DevTools
2. Review nginx error log: `tail -f /var/log/nginx/error.log`
3. Verify certificates: `openssl x509 -noout -text -in cert.crt`
4. Test with curl first: `curl --cert client.p12 https://example.test/`
5. Contact your CA support with full error message